

Statistics and AI driver's license

Using AI (safely) in practice: for learning and for data analysis

Hyungju Jeon · 2026/10/07

Introduction

Using AI in learning/research

What do you want to achieve through a PhD?

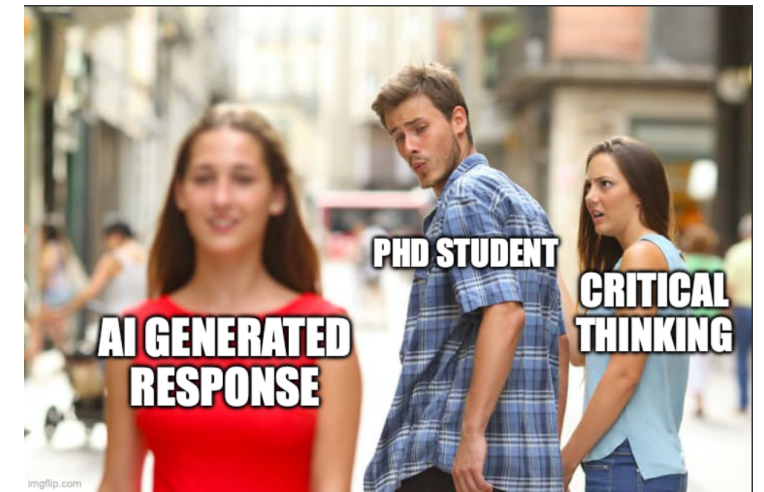
My answer: building a 'taste'. Knowing which questions matter, which results are interesting, and which analyses you can trust.

How can AI help you in achieving this goal?

Understand their strength and weakness, acknowledging risk

The temptation of productivity

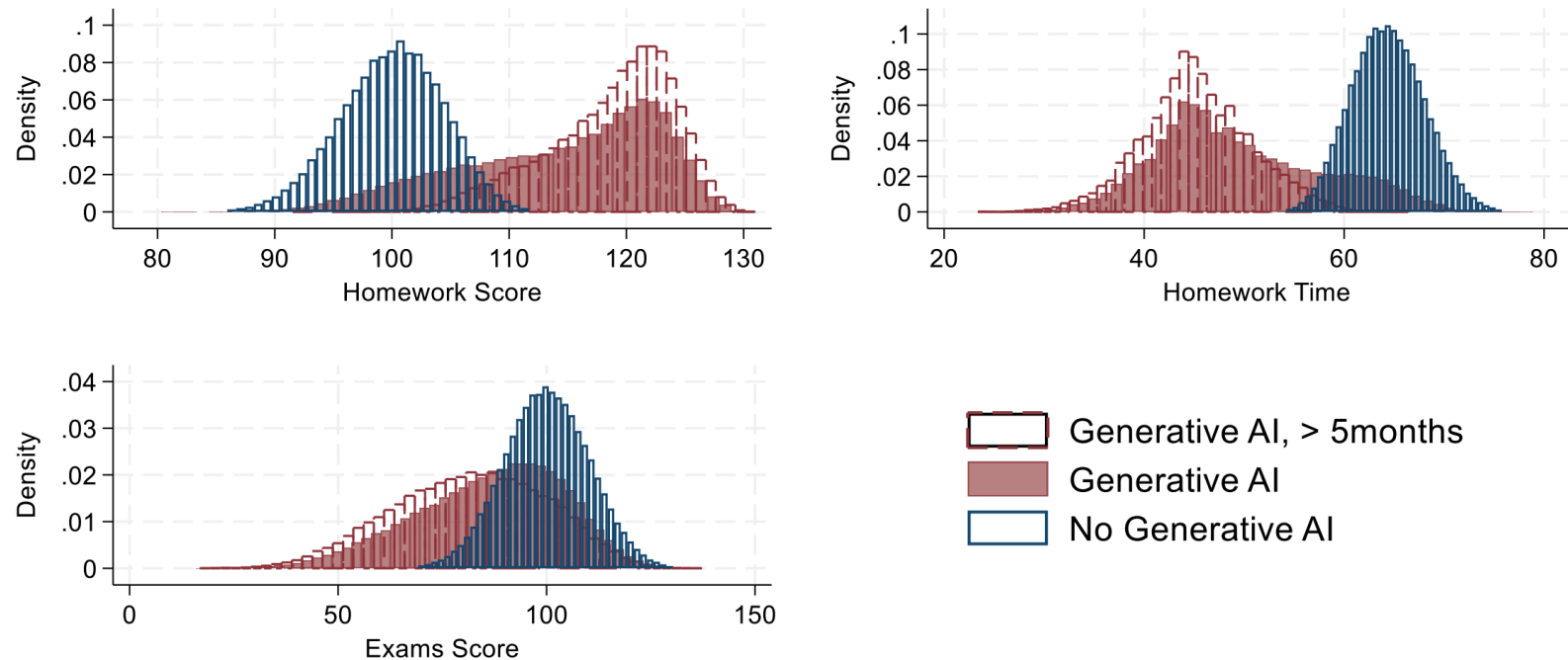
AI lets you produce more, faster: more code, more text, more analyses. It feels like progress, but if the AI does the thinking, you do not build your taste.



Illusion of competence

Reading and understanding AI response $\neq$ Learning

Histograms, GenAI > 5 months



Introduction

Illusion of competence

PNAS

RESEARCH ARTICLE

COMPUTER SCIENCES

OPEN ACCESS



The more you automate, the less you see: Hidden pitfalls of autonomous AI scientists

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Edited by Kenneth W. Wachter, University of California Berkeley, Berkeley, CA; received March 23, 2026; accepted September 2, 2026

The rapid rise of autonomous AI scientists marks a paradigm shift in scientific discovery by automating the research lifecycle. Yet their rushed development has outpaced critical oversight, leaving key workflow decisions dangerously unscrutinized. We present a much-needed systematic analysis of open-source AI scientist systems, investigating four primary pitfalls: inappropriate benchmark selection, data leakage, metric misuse, and post hoc selection bias. Through controlled experiments that isolate each pitfall, we find systematic vulnerabilities across two representative open-source systems. Crucially, we find that these flaws are largely invisible at the level of the final manuscript, suggesting that current manuscript-centric peer review paradigms are fundamentally insufficient for ensuring the integrity of automated research. We further propose mitigation strategies and demonstrate that access to full workflow artifacts (log traces and code) enables more effective auditing. Our findings suggest that journals, conferences, and researchers should move beyond manuscript-only evaluation toward process auditing the end-to-end workflow artifacts of AI scientist systems.

Significance

AI systems are increasingly capable of autonomously conducting scientific research, from generating hypotheses to writing full papers. However, our study finds that these “autonomous AI scientists” can introduce systematic methodological pitfalls such as biased evaluation, undisclosed data manipulation, and selective

[Submitted on 29 Aug 2025]

AI Meets the Learning?

Matthias Lehmann, Philipp B. Cornelius, Fabian J. Sting

The effect of large language models (LLMs) in education is debated: Previous research shows that LLMs can help as well as hurt learning. In two pre-registered and incentivized laboratory experiments, we find no effect of LLMs on overall learning outcomes. In exploratory analyses and a field study, we provide evidence that the effect of LLMs on learning outcomes depends on usage behavior. Students who substitute some of their learning activities with LLMs (e.g., by generating solutions to exercises) increase the volume of topics they can learn about but decrease their understanding of each topic. Students who complement their learning activities with LLMs (e.g., by asking for explanations) do not increase topic volume but do increase their understanding. We also observe that LLMs widen the gap between students with low and high prior knowledge. While LLMs show great potential to improve learning, their use must be tailored to the educational context and students' needs.

RESEARCH ARTICLE

ECONOMIC SCIENCES



Generative AI without guardrails can harm learning: Evidence from high school mathematics

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Hamsa Brunskill, Stanford University, Stanford, CA; received November 3, 2024; accepted May 5, 2025 by Editorial Board Member

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The Generative AI Learning Penalty: Evidence from Chinese Secondary Education

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AI makes you smarter but none the wiser: The disconnect between performance and metacognition

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Using AI to learn

How can we use AI to learn?

Using AI to learn

Ways to learn with AI

Tailor the AI to your goals and to your learning style

Pattern	What it does
Tutor, don't solve	AI explains concept, and asks questions
Three levels	Explain a concept in multi-level abstraction
Interactive guide	Interactive visualization like 3Blue1Brown
AI as a student	Explain a concept in your own words; the AI asks questions and criticizes
AI Review	Point out weakness/errors in my claim but don't fix

And more...

Ask the same question twice

Chat 1: “What is the bootstrap?” Read the answer once.

Chat 2 : “Be my tutor for the bootstrap.

I know the sampling distribution of the mean and the shuffle test.

Do not give me the answer.

Ask me one question at a time, and wait for my answer.”

Close both chats.

Q. What does the bootstrap resample, and what question does it answer that the shuffle test does not?

Prompt patterns

Pattern	Prompt
Tutor, don't solve	"Be my tutor for [topic]. Do not give me the answer. Ask me one question at a time, and wait for my answer."
Three levels	"Explain [concept] for a 5-year-old, an undergraduate, and a PhD student. Keep the facts the same. Tell me what you simplified."
Interactive guide	"Make one self-contained interactive HTML page with sliders that visualizes [concept] as I change [parameter]."
AI as a student	"I will explain [concept] in my own words. Act as a curious student: ask me questions until my explanation has no gaps."
AI Review	"Here is my [claim / code / summary]. Find the weakest point and any errors. Do not fix or rewrite it."

Reading code

Can you explain every line of the code?

Review your morning code

1 • Paste and run

Paste the test you wrote this morning into `code_reading.ipynb`.

2 • Ask for a review

“Do not rewrite or fix it. Say what null hypothesis it tests, then ask me questions.”

3 • Find a bug together

“I ran [this]. I expected [this], but got [that]. Do not fix it. Ask me one question.”

Let it quiz you

“Ask me 3 questions about this code, one at a time.”
Score yourself.

Check your code on a case you know

The coin

9 heads in 10 flips. Two-sided p from the worksheet table?

This morning's tumors

Lean mice, whole mm³: control 543, 83, 555, 483, 557; anti-PD-L1 0, 70, 0, 257, 29. 252 splits. Exact two-sided p ?

Predict first, then run your code.

AI version: change one line, predict, run

Change	Your prediction	Result
alternative="less", then "greater"		
Only 20 shuffles, five different seeds		
Remove the + 1 in the last line		

Write the prediction in your learning log before you run the cell.

If the AI's code is wrong, say exactly how: "I ran [this]. I expected [this], but got [that]. That is wrong because [reason]."

Explain it, then let the AI test you

Three levels — “Explain this code for a 5-year-old, an undergraduate, and a PhD student.”

A visual page — “Make one HTML page that animates one shuffle and builds the histogram.”

Quiz me — the AI asks 3 questions; answer without looking at the code.

AI-assisted learning in action

Different types of statistical errors

How often does the rule accuse an honest dealer?

This morning: a dealer flips a coin; you decide from 10 flips whether it is fair.

Your rule: watch 10 flips; accuse the dealer on 0, 1, 9, or 10 heads.

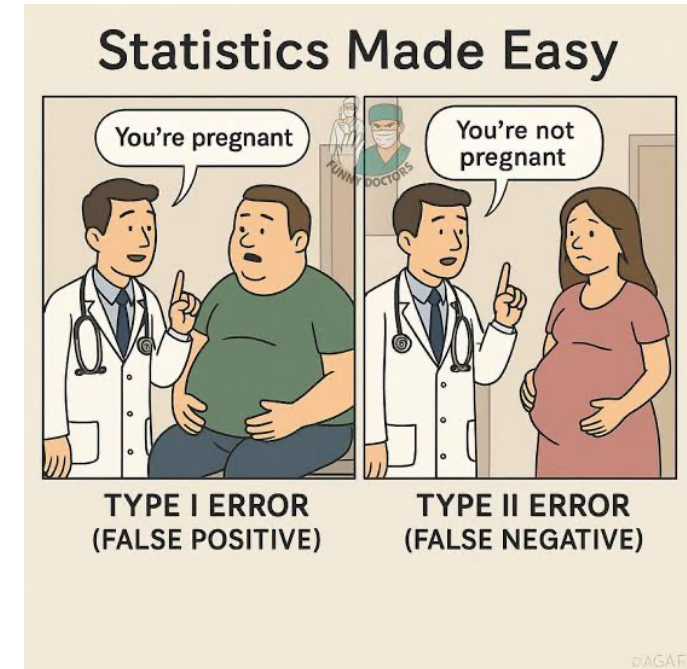
The dealer is honest. How often does your rule accuse them?

Heads in 10 fair flips	0	1	2	3	4	5	6	7	8	9	10
Probability	0.001	0.010	0.044	0.117	0.205	0.246	0.205	0.117	0.044	0.010	0.001

Two types of error

	Dealer is honest	Dealer cheats
Accuse	Type I	
Do not accuse		Type II

1. Which two cells are mistakes?
2. What is the mistake that we computed in the previous slide?
3. How can we compute the other error?



Catching one cheater: the 70% coin

Heads in 10 flips	0–2	3	4	5	6	7	8	9	10
Probability	0.002	0.009	0.037	0.103	0.200	0.267	0.233	0.121	0.028

1. How often does your rule (0, 1, 9, or 10 heads) accuse this cheater?
2. How often does this cheater get away?

Other cheating coins

Coin lands heads	50% (honest)	60%	70%	80%	90%
P(accuse)	0.022		0.15		

Compute the 60%, 80%, and 90% coins. Use $P(9 \text{ heads}) = 10 \cdot p^9 \cdot (1 - p)$ and $P(10 \text{ heads}) = p^{10}$; 0 and 1 heads are tiny.

1. Sketch $P(\text{accuse})$ against how often the coin lands heads. Where does the curve start?
2. Can the Type II error rate of your rule be one number? Write your idea.

AI step

“My rule: flip 10 times, accuse on 0, 1, 9 or 10 heads. The chance of catching a cheater is 0.048 (60% coin), 0.15 (70%), 0.38 (80%), 0.74 (90%). Can its Type II error rate be one number? My idea: [your answer to 2]. Do not give me the answer; ask me one question at a time.”

Reading the power curve

Coin lands heads	50%	60%	70%	80%	90%	100%
P(accuse), 10 flips	0.022	0.048	0.15	0.38	0.74	1

The class's power curve for your rule: 10 flips, accuse on 0, 1, 9, or 10 heads.

1. Where does the curve start, and why there?
2. How biased must the coin be before your rule catches it more often than not?
3. What new idea did the AI tutor give you on the last slide? Share one.

AI step:

“one HTML page that plots the power curve of my rule, with a slider for the number of flips.”

More flips, more power

Flips	Accuse when heads are	False accusations (α)	Catch the 70% cheater (power)
10	≤ 1 or ≥ 9	0.022	
20	≤ 5 or ≥ 15	0.041	
50	≤ 17 or ≥ 33	0.033	
100	≤ 39 or ≥ 61	0.035	

α stays below 0.05 at every size: it is set by the rule. Power grows with the number of flips.

The trade-off between α and β

Accuse when heads are	False accusations (α)	Catch the 70% cheater
0 or 10	0.002	
0, 1, 9 or 10	0.022	
0–2 or 8–10	0.109	

Moving the rule trades α against β .

How can we lower both?

AI step:

“one HTML page with two bar charts on one axis: heads in 10 flips for a fair coin and for a 70% coin. Let me move the rejection region, shade α and β , and show both numbers.”

Explain Type I and Type II errors

In your own words, with the dealer's coin: Type I error, Type II error, the power curve, and the trade-off between α and β .

1. Explain it to the AI.

"I will explain Type I and Type II errors. Do not reply until I write DONE. Then tell me what is right, what is missing, and what is wrong, and ask me one question about the weakest part."

2. Fix the weakest part. Then explain it to your neighbor without notes.

How many accused dealers are cheaters?

	Honest dealers (900)	Cheating dealers (100)
Accused		
Not accused		

In this casino, 1 in 10 dealers cheats with the 70% coin. You watch 10 flips of each of 1,000 dealers and use your rule.

1. Fill in the table. Of the accused dealers, what fraction cheat?
2. The manager says: “We accused 35 of 1,000 dealers. That is 3.5%, less than 0.05, so this casino has no cheating problem.” Is the manager right?
3. Redo 1 with 100 flips per dealer (accuse on ≤ 39 or ≥ 61 heads: $\alpha = 0.035$, power = 0.98), and again for a casino where half the dealers cheat. Which change raises the fraction more?

Estimating the cheat from caught cheaters

Flips	Accuse when heads are	Power (catch the 70% coin)	Estimated heads rate of caught cheaters
10	≤ 1 or ≥ 9	0.15	
20	≤ 5 or ≥ 15	0.42	
100	≤ 39 or ≥ 61	0.98	

All cheaters use the 70% coin. Estimate each caught cheater's coin from its heads (9 of 10 $\rightarrow$ 0.9).

1. Fill in the 10-flip row (you catch only 9 or 10 heads: $P = 0.121, 0.028$).
2. The truth is 0.70. Do caught cheaters look more biased? Predict the other rows.
3. Lab: 5 mice per group, a significant effect. Is the reported effect too big?

Peeking: testing again after more flips

You accuse an honest dealer ...	Probability
after the first 10 flips (≤ 1 or ≥ 9 heads)	0.022
only after all 20 flips (≤ 5 or ≥ 15 heads)	0.033
in total	
if you test only once, after all 20 flips (≤ 5 or ≥ 15 heads)	0.041

Step 1: watch 10 flips; accuse on ≤ 1 or ≥ 9 heads. Step 2: if not accused, watch 10 more; accuse if all 20 flips show ≤ 5 or ≥ 15 heads.

1. The dealer is honest. Can the total chance of an accusation be below 0.022?
2. Add the first two rows. Compare with testing only once, after all 20 flips.
3. Lab: $p = 0.08$ with 5 mice per group, so you add 5 more and test again. Problem?

Power of the shuffle test with few mice

Mice per group	Possible splits	Smallest two-sided p	Can it reach 0.05?
2	6	$2/6 = 0.33$	Never
3	20	$2/20 = 0.10$	Never
4	70	$2/70 \approx 0.029$	Only for the most extreme split
5	252	$2/252 \approx 0.008$	Yes (this morning's tumors)
9	48,620	≈ 0.00004	Yes

Like a coin flipped only a few times: with 3 mice per group, power is 0 whatever the drug does.

Lean vs obese mice: similar effect, different p

This morning's tumors (5 vs 5 mice)	Difference in means	Spread of shuffled differences (SD)	Two-sided p
Lean mice	373 mm ³	158	0.016
Obese mice	295 mm ³	712	0.68

1. Does $p = 0.68$ show that the antibody does nothing in obese mice?
2. What makes a real effect hard to detect here? Which coin result is this like?

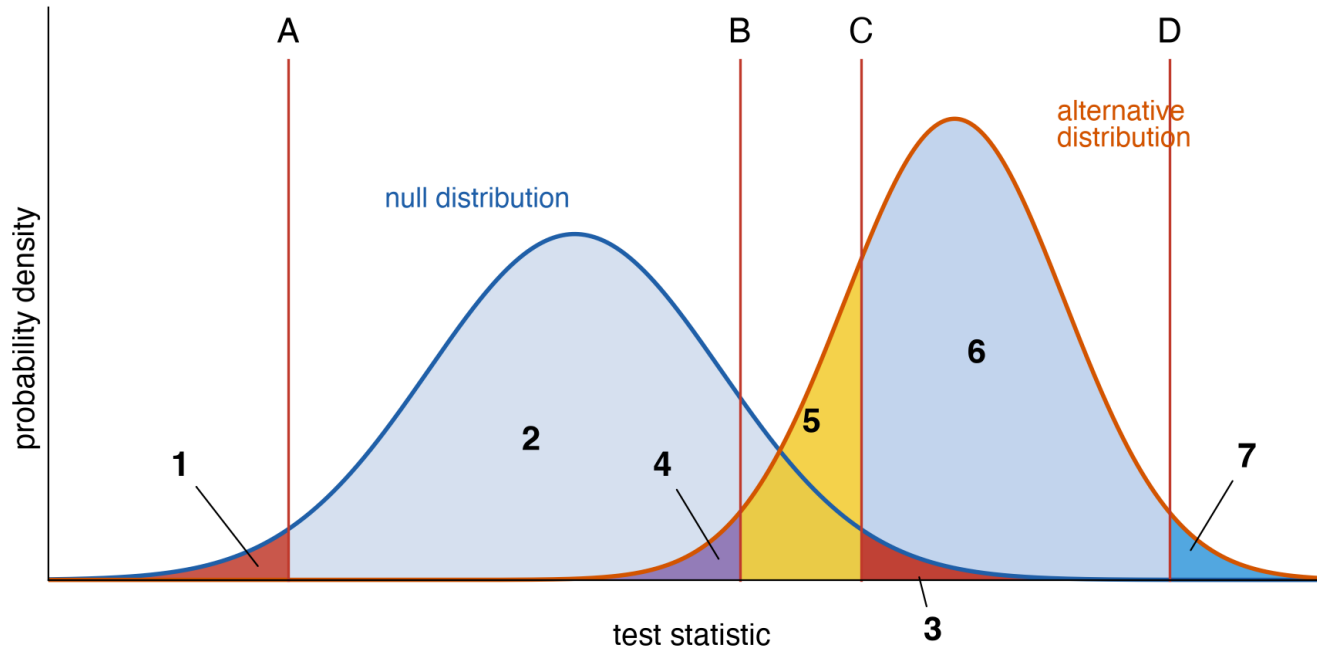
Quiz

What can you do without AI?

15 minutes, pen and paper. No AI, no notes, no notebook.

Write what you think, even if you are not sure. The review is the next step.

Quiz: which area is which?



Lines A and C cut 2.5% tails of the null; B and D cut 2.5% tails of the alternative. $\alpha = 5\%$, two-sided.

1. Which areas add up to α ?
2. Which areas add up to β ?
3. Which areas add up to the power?
4. Which lines does the test use to decide?
5. With more mice, the alternative moves right. Which areas shrink?

Quiz: true or false

True or false? Give a one-line reason for each.

1. The p-value is the chance of data at least this extreme, if the null hypothesis is true.
2. The p-value is the chance that the null hypothesis is true.
3. If $p > \alpha$, the null hypothesis is true.
4. A smaller p-value always means a larger effect.
5. A test has one Type II error rate.
6. With more data and the same α , the power goes up.
7. Counting 15 readings of one mouse as 15 mice keeps false positives near α .
8. Testing again after adding more mice keeps the false-positive rate at α .

Real data

What do real mice say about false positives?

The same questions as the coin, now with 72 real mice: real noise, and a truth you control.

Does memantine change BDNF in the mouse cortex?

Data: BDNF protein in mouse cortex. 72 mice, 15 measurements each, memantine or saline. Open `real_data.ipynb`.

1. The real question: one mean per mouse. Does memantine change BDNF? Predict first: significant or not?
2. Type I with real noise: split saline mice at random into two fake groups. Predict the false-positive rate for one mean per mouse, and for every measurement.
3. Type II with a known effect: add a 1 SD effect to one fake group. Predict the power with 3, 5, and 9 mice per group.

AI step: write two sentences on what the real result means. Ask the AI to attack your weakest claim, then fix it.

Type I error and power with real data

Analysis	Result
Real question: memantine vs saline (one mean per mouse)	$p = 0.29$
Type I, fake groups, one mean per mouse	5.8%
Type I, fake groups, every measurement as if it were a mouse	48.8%
Power at a 1 SD effect, 3 / 5 / 9 mice per group	0.00 / 0.25 / 0.51

$p = 0.29$ is not proof of no effect. Measurements are not mice.

1. Why does testing every measurement raise false positives to almost 50%?
2. $p = 0.29$ with about 20 mice per group: which effects could this study have missed?

Optional • Multiple testing

What if you test 20 dealers?

If time allows.

Testing many honest dealers

Honest dealers tested	Chance that at least one is accused — your prediction
1	
5	
20	

Each honest dealer is accused with probability 0.022 (rule: 0, 1, 9, or 10 heads).

Hint: first find the chance that none is accused.

Design your own correction

In pairs, 5 minutes, AI closed.

Goal: test 20 dealers, and keep the chance of accusing any honest dealer at or below 0.05.

What does your procedure cost? Which cheaters will you now miss?

[Comic suggestion: xkcd 882 'Significant' — green jelly beans and acne.]

Bonferroni: a stricter rule for each dealer

Accuse a dealer when heads are	One honest dealer	Any of 20 honest dealers	Catch one 70% cheater
0, 1, 9 or 10	0.022	0.35	0.15
0 or 10	0.002	0.04	0.03

Dividing the allowed error by the number of tests ($0.05 / 20$) is the idea behind the Bonferroni correction. The price is power.

Then ask the AI to review your own procedure as a sceptic — without giving you its own.

Wrap-up

AI is a powerful tool, if you use it well

AI can be a great tutor.

Beware the illusion of productivity.

Always ask questions.

Find your own way to consolidate.

USES AI AS A TOOL



PRODUCTIVE PhD

COPY/PASTE AI RESPONSE



**DON'T UNDERSTAND
ANYTHING**

“The important thing is not to stop questioning. Curiosity has its own reason for existing.”

— Albert Einstein

Tip: quick side questions without losing your thread: /btw in Claude Code, side chat in Codex.